Compact event and recovery memory: analytical checks

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Scope. Reviewer deductions from the supplied 2006 journal, Section 2.2 and Appendix A. These are not a simulator, a reachable traffic witness, an adopted compact law, or gradient validation. The plain-language report is `event_memory_findings_and_gate.md`; source locations are in `event_memory_source_audit.md`. Earlier mathematical artifacts are unchanged. The selected reference uses sampled descriptor births.

1. Threshold progress is exact event bookkeeping

Let e denote one source descriptor-generating transfer stream (origin cell, origin lane, target lane). At consecutive source sampling times, g_e^k is its cumulative actual transferred vehicle-equivalent mass. Let $m_* > 0$ be the source threshold unit, $\Delta n_e^k \geq 0$ the actual mass transferred between these samples, and z_e^k its dimensionless remainder. Then

$$g_e^{k+1} = g_e^k + \Delta n_e^k, \quad z_e^k = \frac{g_e^k}{m_*} - \left\lfloor \frac{g_e^k}{m_*} \right\rfloor, \quad 0 \leq z_e^k < 1. \tag{1}$$

The new descriptor count b_e^k and next remainder satisfy

$$b_e^k = \left\lfloor \frac{g_e^{k+1}}{m_*} \right\rfloor - \left\lfloor \frac{g_e^k}{m_*} \right\rfloor = \left\lfloor z_e^k + \frac{\Delta n_e^k}{m_*} \right\rfloor, \qquad z_e^{k+1} = z_e^k + \frac{\Delta n_e^k}{m_*} - b_e^k. \tag{2}$$

The identity follows by removing the integer part of g_e^k/m_*. It handles more than one threshold crossing in a sample. No within-sample birth interpolation is added. Use the same placement of sampling times as the source's cumulative-transfer convention, not a new event scheduler.

All traffic mass Δn_e^k remains in the paired donor/receiver balance. The descriptor count is metadata. Zero new births does not imply zero old restrictions, and a descriptor's existence does not imply that it binds. Store a remainder for every generating stream, not just a single ramp total. The result is conditional on given own-model transfers; it does not force two independently evolving models to produce identical births.

2. Source-sampled free motion admits age-based reconstruction

On a level road, the supplied acceleration is $a(v) = a_0(1 - v/v_{\max})$, with $a_0 > 0$ (m/s^2) and vehicle parameter $v_{\max} > 0$ (m/s). The KW free speed u is a different quantity. Consider only the subcase in which downstream traffic offers speed u throughout a restriction's path, $0 \le v_b < u < v_{\max}$, and $r = 1 - a_0\Delta t/v_{\max} \in (0, 1)$. Let v_n be realized speed after n source motion steps since birth. The supplied desired-speed and downstream-cap operations reduce to

$$v_{n+1} = \min\{u, rv_n + a_0\Delta t\}, \qquad v_0 = v_b. \tag{3}$$

Induction gives

$$v_n = \min\{u, v_{\max} - (v_{\max} - v_b)r^n\}, \qquad x_n = x_b + \Delta t \sum_{j=0}^{n-1} v_j. \tag{4}$$

The position sum uses the supplied current-speed update, not the next speed. For the induction, the affine map is increasing and maps u to a value above u. Once clipped to u, the subsequent clipped value remains u. The first source step attaining u is

$$n_* = \left\lceil \frac{\log((v_{\max} - u)/(v_{\max} - v_b))}{\log r} \right\rceil. \tag{5}$$

The continuous free-motion calculation is only a diagnostic limit:

$$V(s) = v_{\max} - (v_{\max} - v_b)e^{-a_0 s/v_{\max}}, \qquad \tau_u = \frac{v_{\max}}{a_0} \log \frac{v_{\max} - v_b}{v_{\max} - u}. \tag{6}$$

It is not substituted for the reference's sampled update. If $u = v_{\max}$, finite exact attainment is not supplied by that expression; this boundary must not be hidden by a timeout or tolerance.

Birth-step, position and speed records therefore reconstruct free kinetics, and groups with identical birth conditions share a path. Under general traffic, the realized speed uses a local downstream cap and the next desired speed starts from that realized speed. A free birth-age clock does not encode a later slowdown or its changed position. No transient flow/service map follows from this kinematic reconstruction alone.

3. Count and mean speed do not generally close the capped update

This is a local algebraic test, not a constructed freeway experiment. Write the level-road desired update as $Y = \phi(v) = b + cv$, with $b = a_0\Delta t > 0$, $c = r \in (0, 1)$. With a common next-step downstream cap d, the realized next speed is $\min(d, Y)$. Choose symbolic $\varepsilon > 0$ such that $b < d - 2\varepsilon$ and $(d + 2\varepsilon - b)/c \le u$. Two possible input lists have desired speeds

$$(Y_1, Y_2) = (d - \varepsilon, d + \varepsilon), \qquad (\widetilde{Y}_1, \widetilde{Y}_2) = (d - 2\varepsilon, d + 2\varepsilon). \tag{7}$$

Their current speed lists are $(Y_i - b)/c$, so both have count two and the same current speed sum $2(d - b)/c$. However,

$$\sum_{i=1}^{2} \min(d, Y_i) = 2d - \varepsilon, \qquad \sum_{i=1}^{2} \min(d, \widetilde{Y}_i) = 2d - 2\varepsilon. \tag{8}$$

For any finite desired-speed counting measure μ, the missing quantity is visible through

$$\int \min(d, y)\, \mu(dy) = \int y\, \mu(dy) - \int (y - d)_+\, \mu(dy). \tag{9}$$

Knowing the first integral alone does not determine the cap-dependent tail. At a cap with no atom, the derivative of the left side with respect to d equals $\mu((d, \infty))$. A changing cap can query different parts of the distribution. Without capping, the affine sum would be closed by count and speed sum.

Limits. This rules out that particular exact moment update over the displayed admissible local inputs. It does not establish their reachability under a common metering schedule, identical observed histories, a TST effect, or the impossibility of every finite summary. The relevant task may have an invariant restricted distribution or a different sufficient aggregate closure.

4. An unlabelled configuration is an exact endpoint, not compression

Let $\mathcal{K}$ denote the ordinary continuous traffic field, waiting and any cumulative-count state required by the source solver. For lane ℓ, let η_ℓ record retained descriptors with position x_i and realized speed v_i at a consistently chosen source step:

$$\eta_\ell = \sum_{i:\ell_i=\ell} \delta_{(x_i,v_i)}, \qquad \eta_\ell(\mathbb{R}\times[0,u]) = \text{descriptor count, not traffic mass.} \tag{10}$$

If type parameters differ, include their fixed marks; no heterogeneity is introduced by this notation. Given a well-defined source update depending on physical positions/speeds and fixed structural marks, but not arbitrary names, its action on the measure can be written schematically as

$$\eta_\ell^{k+1} = (\Phi_{\mathcal{K}^k,\eta^k})_{\#}\eta_\ell^k + \mathcal{B}_\ell^k - \mathcal{D}_\ell^k. \tag{11}$$

Φ denotes the source motion map, # its push-forward (moving every record according to that map), $\mathcal{B}$ sampled own-transfer births, and $\mathcal{D}$ exits or removals proved permanently redundant under the declared domain. A temporary nonbinding observation is not by itself a permanent-removal proof. This is a representation identity, not a new PDE, histogram transport rule, or certified joined solver.

Permuting descriptor names leaves η unchanged and, under that coupling assumption, leaves physical constraints unchanged. Positions still encode spatial order. At atomic resolution this retains one record per descriptor, so it does not demonstrate a small fixed-dimensional memory.

A grouped position/speed representation needs a declared projection, independent update and service law. Averaging trajectories, spreading a point restriction over a whole bin, or summing separate capacity penalties is not entailed by (10)–(11). None is adopted here. The question about this broader state family is the next user discussion, S3-G2-Q-M.

Overall result. Event progress is exact; free kinetic memory is conditionally supported; one general clock or count/mean-speed update is not qualified. A genuinely compressed source-consistent recovery/service law, joined events/clearance, complete TST response and all gradient claims remain open. Stage 4 and experiments remain unopened.